INVERTIBLE EQUALS LOCALLY FREE OF RANK ONE WHEN THE LOCAL RINGS ARE LOCAL

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The characterisation of invertible modules through the evaluation map into the structure sheaf, and the local splitting off a finite free module. Invoked inside the counted proof (source lines 4400-4401 and 4405-4406) as pointers to a lemma of this chapter. Provides the trivialisation test in one direction and the local freeness input in the other.
- The classification of invertible modules over a local ring, and the criterion for a finitely presented module with free stalks to be locally free of that rank. Both are invoked inside the counted proof (source lines 4408-4414) as a pointer to a section of another chapter and to a lemma of this chapter. Upgrades freeness of each stalk over a local ring to local freeness of the sheaf.

Source reference. Original lemma lemma-invertible: sources/stacks-modules/pinned-original-source.tex, source lines 4367–4367. Navigation pointer only: the counted text refers to the Stacks item lemma-invertible, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference more-algebra-section-picard: sources/stacks-modules/pinned-original-source.tex, source lines 4260–4260. Navigation pointer only: the counted text refers to the Stacks item more-algebra-section-picard, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

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Source reference. Original lemma lemma-finite-presentation-stalk-free: sources/stacks-modules/pinned-original-source.tex, source lines 2407–2407. Navigation pointer only: the counted text refers to the Stacks item lemma-finite-presentation-stalk-free, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $(X, \mathcal{O}_X)$ be a ringed space. Any locally free $\mathcal{O}_X$-module of rank 1 is invertible. If all stalks $\mathcal{O}_{X,x}$ are local rings, then the converse holds as well (but in general this is not the case).*

Proof. The parenthetical statement follows by considering a one point space X with sheaf of rings $\mathcal{O}_X$ given by a ring R. Then invertible $\mathcal{O}_X$-modules correspond to invertible R-modules, hence as soon as $\mathrm{Pic}(R)$ is not the trivial group, then we get an example.

Assume $\mathcal{L}$ is locally free of rank 1 and consider the evaluation map

$$\mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{H}om_{\mathcal{O}_X}(\mathcal{L}, \mathcal{O}_X) \longrightarrow \mathcal{O}_X$$

Looking over an open covering trivialization $\mathcal{L}$, we see that this map is an isomorphism. Hence $\mathcal{L}$ is invertible by Lemma lemma-invertible.

Assume all stalks $\mathcal{O}_{X,x}$ are local rings and $\mathcal{L}$ invertible. In the proof of Lemma lemma-invertible we have seen that $\mathcal{L}_x$ is an invertible $\mathcal{O}_{X,x}$-module for all $x \in X$. Since $\mathcal{O}_{X,x}$ is local, we see that $\mathcal{L}_x \cong \mathcal{O}_{X,x}$ (More on Algebra, Section more-algebra-section-picard). Since $\mathcal{L}$ is of finite presentation by Lemma lemma-invertible we conclude that $\mathcal{L}$ is locally free of rank 1 by Lemma lemma-finite-presentation-stalk-free. □